PROBLEM 32

Rheonomic Spool Constraint

Particle with String on Spool

Original statement · Problems.pdf, p. 20

A heavy particle P of mass m is connected by a string of total length $l = 4\pi a$ to a cylindrical spool of radius a and center at O. The spool rotates at a known constant angular rate ω counterclockwise. The string is massless, inelastic, thin, and remains always in tension during the motion of the particle. We define the inertial reference frame Oxy. The other end of the string is attached to the spool at a point A of its perimeter, and we shall call B the instantaneous point of tangency between the straight segment of the string (the segment BP) and the spool. We will denote the angle between the local horizontal (axis Ox) and the radius OB as ϕ. Initially, $\mathbf{OA} = -a\mathbf{i}$, and B never reaches A.

(a) Discuss the number of degrees of freedom of the motion of the particle.

(b) Write down the position, velocity, and acceleration vectors of P with respect to Oxy as a function of the constants of the problem, time, the angle ϕ, and its derivatives.

(c) Write down the mathematical expression of the forces acting on P.

(d) Is the mechanical energy of P conserved? If so, write down the equation of conservation of mechanical energy; otherwise, explain why it is not conserved.

(e) Write down a closed set of equations that, if solved, would yield ϕ (and the other unknowns of the problem, if any) as a function of time. Note that you do not need to solve the equations.

(f) Consider now that $\omega = 0$. Can you write down a first integral of motion that only involves ϕ and $\dot{\phi}$ (but not $\ddot{\phi}$) in this case? You may want to rethink your previous answers for this particular case.

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** A particle P of mass m is attached to a string of length $l = 4\pi a$ unwinding from a circular spool of radius a rotating counterclockwise at constant rate ω.
- **Initial Tangency:** Initially at $t = 0$, A is at $(-a, 0)$ and the string is wrapped around the spool.
- **CDOFs:** 1 degree of freedom: the instantaneous unwinding angle $\phi(t)$.
- **Parameters:** $m, a, l = 4\pi a, \omega, g > 0$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- Inertial frame: Oxy.
- Position of contact point B: $\mathbf{r}_B = a\cos\phi\,\mathbf{i} + a\sin\phi\,\mathbf{j}$.
- Unit vectors at B:
- Radial: $\mathbf{u}_r = \cos\phi\,\mathbf{i} + \sin\phi\,\mathbf{j}$.
- Tangential: $\mathbf{u}_t = -\sin\phi\,\mathbf{i} + \cos\phi\,\mathbf{j}$ (counterclockwise tangent; the straight segment leaves B clockwise, towards the lower right in the figure, i.e. along $-\mathbf{u}_t$).
- Straight segment length:

The initial wrapped length was l. As the spool rotates at rate ω, the length of the unwound free segment BP is:

$$s(t) = a(3\pi - \omega t + \phi)$$

- Position of particle P:

$$\mathbf{r}_P = \mathbf{r}_B - s(t)\mathbf{u}_t = a\cos\phi\,\mathbf{i} + a\sin\phi\,\mathbf{j} - a(3\pi - \omega t + \phi)(-\sin\phi\,\mathbf{i} + \cos\phi\,\mathbf{j})$$

PHASE 3

Step-by-Step Mathematical Deduction

Part (a): Degrees of Freedom

- Although P moves in the plane (2 coordinates), the constraint of remaining tangent to the unwinding spool leaves exactly **1 degree of freedom** (ϕ).

Part (b): Velocity and Acceleration Vectors

Differentiating $\mathbf{r}_P(t)$ with respect to time:

$$\mathbf{v}_P = \frac{d\mathbf{r}_P}{dt} = a\dot{\phi}\mathbf{u}_t - \dot{s}\mathbf{u}_t + s\dot{\phi}\mathbf{u}_r$$

Since $s(t) = a(3\pi - \omega t + \phi) \implies \dot{s} = a(\dot{\phi} - \omega)$:

$$\mathbf{v}_P = a\omega\,\mathbf{u}_t + s\dot{\phi}\mathbf{u}_r$$

Differentiating once more, with $\dot{\mathbf{u}}_t = -\dot{\phi}\mathbf{u}_r$ and $\dot{\mathbf{u}}_r = \dot{\phi}\mathbf{u}_t$:

$$\mathbf{a}_P = \left[s\ddot{\phi} + a\dot{\phi}^2 - 2a\omega\dot{\phi}\right]\mathbf{u}_r + s\dot{\phi}^2\mathbf{u}_t$$

Part (c): Forces Acting on P

1. Gravity: $\mathbf{F}_g = -mg\mathbf{j} = -mg(\sin\phi\,\mathbf{u}_r + \cos\phi\,\mathbf{u}_t)$.

2. Tension: $\mathbf{T} = +T\mathbf{u}_t$ ($T > 0$ directed towards B, since $\mathbf{r}_B - \mathbf{r}_P = s\,\mathbf{u}_t$).

> **Erratum:** The first version placed the particle at $\mathbf{r}_B + s\mathbf{u}_t$ (up and to the left of B), which contradicts the figure where the segment BP leaves B towards the lower right. Correct: $\mathbf{r}_P = \mathbf{r}_B - s\mathbf{u}_t$, $\mathbf{v}_P = a\omega\,\mathbf{u}_t + s\dot\phi\mathbf{u}_r$, $\mathbf{a}_P = [s\ddot\phi + a\dot\phi^2 - 2a\omega\dot\phi]\mathbf{u}_r + s\dot\phi^2\mathbf{u}_t$ and tension $+T\mathbf{u}_t$; the equations of part (e) and the energy of part (f) were already consistent with this. Numeric check: sympy differentiation of $\mathbf{r}_P$ reproduces $\mathbf{v}_P$ and $\mathbf{a}_P$, and RK4 of (e) with $\omega = 0$ conserves the energy of (f) to 10^{-13} with $T > 0$.

Part (d): Mechanical Energy Discussion

For $\omega \neq 0$, the constraint is **rheonomic** (explicitly time-dependent). The contact point B moves with velocity $\mathbf{v}_B = a\omega\mathbf{u}_t$, so the string tension does work on P. Mechanical energy is **not conserved** for $\omega \neq 0$.

Part (e): Closed Set of Equations

Projecting along the tangent $\mathbf{u}_t$ and normal $\mathbf{u}_r$:

$$a\left[(3\pi - \omega t + \phi)\ddot\phi - 2\omega\dot\phi + \dot\phi^2\right] = -g\sin\phi$$

$$T = ma(3\pi - \omega t + \phi)\dot\phi^2 + mg\cos\phi$$

Compare with the official solution key (Problems.pdf, Solution keys section).

Part (f): Case $\omega = 0$ (Static Spool)

When $\omega = 0$, the constraint is scleronomic and the string does zero work. Mechanical energy is strictly conserved:

$$E = \frac{1}{2}m[a(3\pi + \phi)]^2\dot\phi^2 + mg\left[a\sin\phi - a(3\pi + \phi)\cos\phi\right] = \text{constant}$$

Compare with the official solution key (Problems.pdf, Solution keys section).

PHASE 4

Physical Interpretation & Dimensional Verification

- The system behaves as an involute pendulum whose effective string length increases or decreases linearly as it unwinds.

Source: Problems.pdf (Page 20)